

BANANA LEAF DISEASE DETECTION AND CLASSIFICATION

Computer Vision Pipeline using HOG + SVM and MobileNetV2

1. Introduction

Banana leaf diseases can reduce crop quality and yield. This project uses computer vision and machine learning to detect and classify different banana leaf diseases from images.

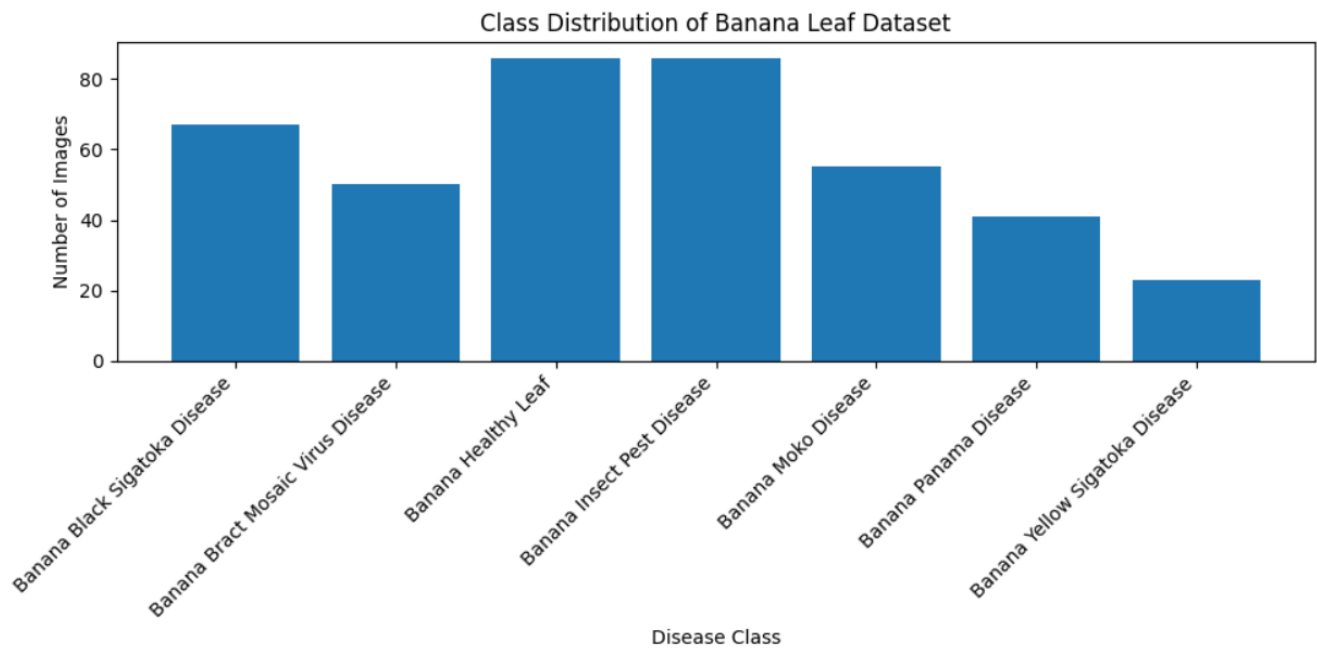
2. Dataset Understanding

The dataset contains 408 images belonging to 7 different banana leaf disease classes, including healthy leaves and diseased leaves.

2.1 Class Distribution

The dataset is imbalanced, with Banana Healthy Leaf and Banana Insect Pest Disease having the most images, while Banana Yellow Sigatoka Disease has the fewest.

Class	Number of Images
Banana Healthy Leaf	86
Banana Insect Pest Disease	86
Banana Black Sigatoka Disease	67
Banana Moko Disease	55
Banana Bract Mosaic Virus Disease	50
Banana Panama Disease	41
Banana Yellow Sigatoka Disease	23



2.2 Data Outliers

The images were checked for corrupted or unreadable files, and no corrupted images were found in the dataset.

2.3 Dataset Bias

The dataset may contain bias due to differences in image background, lighting, leaf orientation, and the unequal number of images in each class. The smallest class has only 23 images, so performance for this class can be less stable.



3. Data Preprocessing and Dataset Split

For HOG + SVM, images were resized to 128×128 pixels before feature extraction. For MobileNetV2, images were resized to 224×224 pixels and the MobileNetV2 preprocessing function was applied.

A stratified split was used to maintain class proportions across the training, validation, and test sets.

Dataset Split	Number of Images
Training	285
Validation	61
Testing	62

4. Model Selection

Two approaches were selected to compare a classical machine-learning method with a lightweight deep-learning method: (1) HOG + Support Vector Machine (SVM), and (2) MobileNetV2 transfer learning.

4.1 HOG + SVM

HOG extracts important shape and edge features from leaf images, while SVM uses these features to classify the disease. It was selected as a classical machine-learning baseline.

4.2 MobileNetV2

MobileNetV2 is a lightweight convolutional neural network that can learn useful visual patterns from images. Transfer learning was selected because the dataset is relatively small. Class weights were used during training to reduce the effect of class imbalance.

5. Evaluation Metrics

Accuracy, precision, recall, and F1-score were used to evaluate the models. Accuracy measures overall correct predictions. Precision measures the correctness of positive class predictions, while recall measures how many actual samples of a class were detected. F1-score balances precision and recall.

Because the dataset is imbalanced, macro F1-score is particularly useful because it gives equal importance to each class.

6. Results

6.1 HOG + SVM Results

The HOG + SVM model achieved a test accuracy of 50.00%. It performed well for Healthy Leaf and Bract Mosaic Virus Disease, but showed poor performance for several other disease classes.

HOG + SVM accuracy and classification report:

```
from sklearn.metrics import accuracy_score, classification_report

# Create and train SVM
svm_model = SVC(kernel="rbf", probability=True, random_state=42)

svm_model.fit(X_train_hog, y_train)

# Predict on test data
y_pred_svm = svm_model.predict(X_test_hog)

# Accuracy
svm_accuracy = accuracy_score(y_test, y_pred_svm)

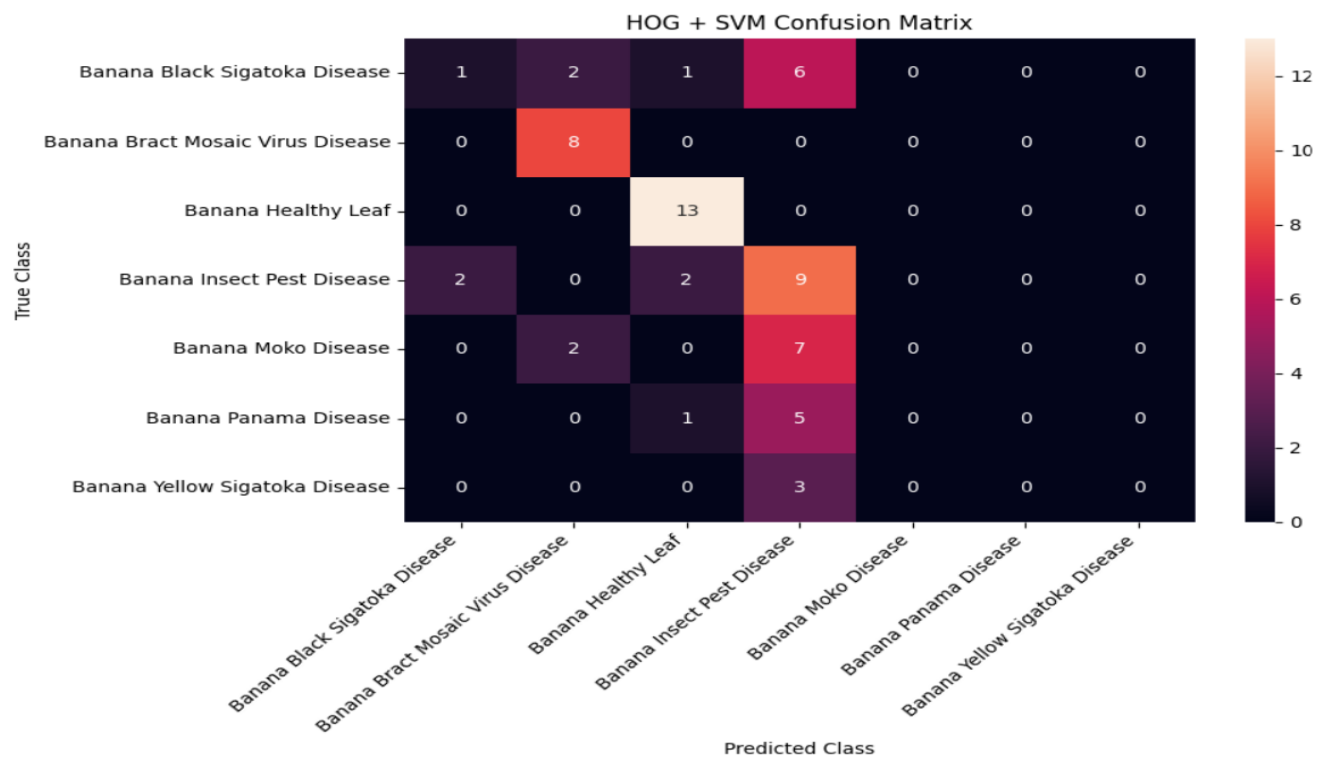
print("SVM Test Accuracy:", round(svm_accuracy * 100, 2), "%")
print("\nClassification Report:")
print(classification_report(y_test, y_pred_svm))
```

SVM Test Accuracy: 50.0 %

Classification Report:

	precision	recall	f1-score	support
Banana Black Sigatoka Disease	0.33	0.10	0.15	10
Banana Bract Mosaic Virus Disease	0.67	1.00	0.80	8
Banana Healthy Leaf	0.76	1.00	0.87	13
Banana Insect Pest Disease	0.30	0.69	0.42	13
Banana Moko Disease	0.00	0.00	0.00	9
Banana Panama Disease	0.00	0.00	0.00	6
Banana Yellow Sigatoka Disease	0.00	0.00	0.00	3
accuracy			0.50	62
macro avg	0.29	0.40	0.32	62
weighted avg	0.36	0.50	0.40	62

HOG + SVM confusion matrix:



6.2 MobileNetV2 Results

The MobileNetV2 model achieved a test accuracy of 90.32% and a macro F1-score of 0.87. It showed substantially better performance across the disease classes.

MobileNetV2 training output:

```
Epoch 1/15
18/18 ----- 21s 840ms/step - accuracy: 0.4632 - loss: 1.6182 - val_accuracy: 0.7213 - val_loss: 0.8780
Epoch 2/15
18/18 ----- 14s 787ms/step - accuracy: 0.7684 - loss: 0.7595 - val_accuracy: 0.8689 - val_loss: 0.4844
Epoch 3/15
18/18 ----- 13s 687ms/step - accuracy: 0.8912 - loss: 0.4385 - val_accuracy: 0.9016 - val_loss: 0.3934
Epoch 4/15
18/18 ----- 12s 679ms/step - accuracy: 0.9158 - loss: 0.2885 - val_accuracy: 0.9016 - val_loss: 0.3342
Epoch 5/15
18/18 ----- 14s 784ms/step - accuracy: 0.9474 - loss: 0.2081 - val_accuracy: 0.8852 - val_loss: 0.2849
Epoch 6/15
18/18 ----- 12s 650ms/step - accuracy: 0.9754 - loss: 0.1787 - val_accuracy: 0.9016 - val_loss: 0.2921
Epoch 7/15
18/18 ----- 20s 598ms/step - accuracy: 0.9719 - loss: 0.1455 - val_accuracy: 0.9016 - val_loss: 0.2734
Epoch 8/15
18/18 ----- 12s 650ms/step - accuracy: 0.9754 - loss: 0.1308 - val_accuracy: 0.8852 - val_loss: 0.2687
Epoch 9/15
18/18 ----- 13s 706ms/step - accuracy: 0.9825 - loss: 0.0935 - val_accuracy: 0.9016 - val_loss: 0.2547
Epoch 10/15
18/18 ----- 19s 648ms/step - accuracy: 0.9825 - loss: 0.0935 - val_accuracy: 0.9180 - val_loss: 0.2374
Epoch 11/15
18/18 ----- 21s 623ms/step - accuracy: 0.9825 - loss: 0.0881 - val_accuracy: 0.9016 - val_loss: 0.2421
Epoch 12/15
18/18 ----- 13s 648ms/step - accuracy: 0.9965 - loss: 0.0731 - val_accuracy: 0.9180 - val_loss: 0.2244
Epoch 13/15
18/18 ----- 12s 607ms/step - accuracy: 0.9825 - loss: 0.0776 - val_accuracy: 0.9344 - val_loss: 0.2199
Epoch 14/15
18/18 ----- 21s 695ms/step - accuracy: 0.9930 - loss: 0.0549 - val_accuracy: 0.9180 - val_loss: 0.2087
Epoch 15/15
18/18 ----- 20s 658ms/step - accuracy: 0.9895 - loss: 0.0559 - val_accuracy: 0.9180 - val_loss: 0.2070
```

MobileNetV2 accuracy and classification report:

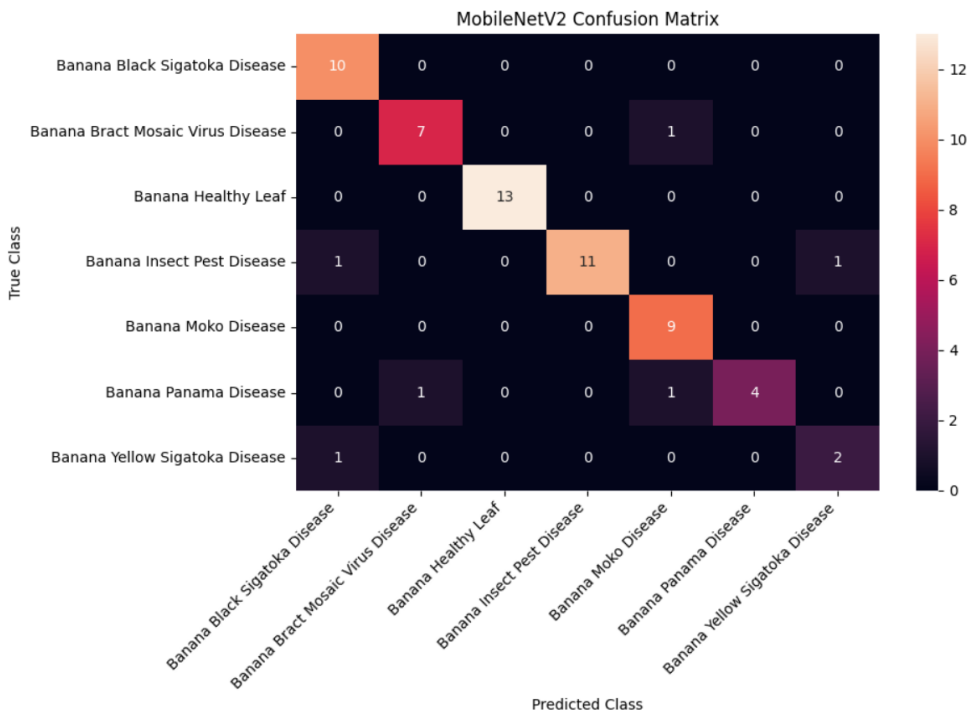
4/4 5s 814ms/step

MobileNetV2 Test Accuracy: 90.32 %

Classification Report:

	precision	recall	f1-score	support
Banana Black Sigatoka Disease	0.83	1.00	0.91	10
Banana Bract Mosaic Virus Disease	0.88	0.88	0.88	8
Banana Healthy Leaf	1.00	1.00	1.00	13
Banana Insect Pest Disease	1.00	0.85	0.92	13
Banana Moko Disease	0.82	1.00	0.90	9
Banana Panama Disease	1.00	0.67	0.80	6
Banana Yellow Sigatoka Disease	0.67	0.67	0.67	3
accuracy			0.90	62
macro avg	0.88	0.86	0.87	62
weighted avg	0.91	0.90	0.90	62

MobileNetV2 confusion matrix:



6.3 Model Comparison

Model	Test Accuracy	Macro F1
HOG + SVM	50.00%	0.32
MobileNetV2	90.32%	0.87

MobileNetV2 outperformed HOG + SVM by a large margin. This suggests that learned deep visual features were more effective than handcrafted HOG features for distinguishing disease patterns in this dataset.

7. Example Model Prediction

One test image was selected to demonstrate an individual prediction. The image belonged to the Banana Bract Mosaic Virus Disease class. MobileNetV2 correctly predicted the same class with a confidence of 84.49%.

Example prediction:

True: Banana Bract Mosaic Virus Disease
Predicted: Banana Bract Mosaic Virus Disease
Confidence: 84.49%



8. Limitations

- The dataset is relatively small, containing only 408 images.
- The class distribution is imbalanced, with only 23 images in the smallest class.
- The test set contains only 3 images for Banana Yellow Sigatoka Disease, so its metrics are less stable.
- Variations in background, lighting, leaf orientation, and image quality may affect model performance.
- The training accuracy of MobileNetV2 became much higher than validation accuracy, indicating some degree of overfitting.

9. Conclusion

Two computer-vision approaches were implemented and compared for banana leaf disease classification. HOG + SVM achieved 50.00% test accuracy, while MobileNetV2 achieved 90.32% test accuracy with a macro F1-score of 0.87.

Based on the results, MobileNetV2 was the better-performing approach for this dataset. The experiment demonstrates that transfer learning with a lightweight CNN can provide substantially stronger disease classification performance than handcrafted HOG features with an SVM baseline.

10. Overall Pipeline

Dataset → Data Inspection → Preprocessing → Stratified Split → HOG + SVM / MobileNetV2 → Test Evaluation → Model Comparison → Example Prediction

